

Scenario-Based SQL Assignment — BikeStores

Assessment	Scenario-Based SQL Assignment
Duration	90 minutes
Total Marks	50 marks
Dataset	BikeStores sample database (SQL Server)
Submission	One .sql file containing clearly numbered solutions

Student Name:

Student ID:

Batch:

Date:

Instructions

- Use the standard BikeStores database and Microsoft SQL Server syntax.
- Write executable SQL for every task. Add brief comments to identify each task.
- Do not modify the source dataset unless a task explicitly asks you to write a data-modification statement.
- Where a business metric is requested, use $\text{quantity} \times \text{list_price} \times (1 - \text{discount})$ as net sales/revenue.
- Readable, logically structured SQL is part of the assessment.
- You may use aliases and intermediate query structures where appropriate. Choose the SQL technique yourself unless a task explicitly requires an object or feature.

Business Scenario

You have joined BikeStores as a Junior Data Engineer. Management wants a Sales & Inventory Intelligence layer that helps store managers understand revenue, customer behavior, product performance, and inventory risk. The operational BikeStores database already contains customers, orders, order items, products, categories, brands, stores, staff, and stock levels. Your job is to answer the following business requirements using SQL.

Useful Schema Reference

Area	Tables	Typical Relationships
Sales	sales.customers, sales.orders, sales.order_items	customer_id, order_id

Organization	sales.stores, sales.staffs	store_id, staff_id
Products	production.products, production.categories, production.brands	product_id, category_id, brand_id
Inventory	production.stocks	store_id, product_id

Assignment Tasks

Task 1 — Build the Sales Detail Dataset (6 marks)

Management needs a detailed sales dataset for analysis. Return one row per order item containing:

- order_id and order_date
- customer full name
- store name
- staff full name
- product name
- category name
- brand name
- quantity, list_price, discount
- calculated net_line_revenue

Include only completed orders (order_status = 4). Sort the result from newest order to oldest.

Task 2 — Store Performance Summary (5 marks)

Create a store-level performance report for completed orders showing:

- store name
- number of distinct orders
- total units sold
- total net revenue
- average order value

Return one row per store and order the stores from highest to lowest total net revenue.

Task 3 — High-Value Customers (5 marks)

Management wants to identify high-value customers. Return customers whose total completed-order spending is greater than the average total spending of customers who have completed orders.

Show customer_id, customer name, completed order count, and total spending. Order the result by total spending descending.

Task 4 — Inventory Risk Report (5 marks)

Operations wants to identify inventory risk. Return products where the stock quantity is below 5 in at least one store.

Show product name, store name, current quantity, category name, and brand name. Products with zero stock should appear first, followed by the lowest remaining quantities.

Task 5 — Top Products Within Each Category (6 marks)

For each product category, identify the top 3 products by total net revenue from completed orders.

Return category name, product name, total units sold, total net revenue, and the product's position within its category. Tied products must receive the same position and the next position should not contain gaps.

Task 6 — Monthly Sales Trend (6 marks)

Create a monthly sales trend for completed orders.

For each calendar month return:

- year
- month
- total net revenue
- previous month's total net revenue
- revenue change from the previous month

The first month may have NULL for the previous-month comparison. Sort chronologically.

Task 7 — Reusable Reporting View (4 marks)

Create a view named `sales.vw_customer_sales_summary` that returns one row per customer and includes:

- `customer_id`
- customer full name
- total number of completed orders
- total units purchased
- total net revenue
- most recent completed order date

Customers with no completed orders must still be represented where possible, with appropriate zero/NULL values.

Task 8 — Safe Data Modification (4 marks)

A customer with `customer_id = 1` has requested that their phone number be changed to '(999) 555-0101'.

Write SQL that performs this update inside an explicit transaction. Include a validation query after the UPDATE and show how the change can be rolled back during testing so the assessment database is not permanently changed.

Task 9 — Store Sales Procedure (6 marks)

Create a stored procedure named `sales.usp_store_sales_report` with these input parameters:

- `@store_id`
- `@start_date`
- `@end_date`

The procedure should return completed-order sales for the requested store and date range, grouped by product. Return product name, total units sold, and total net revenue, ordered by revenue descending.

Add appropriate error handling for invalid date ranges where @start_date is later than @end_date.

Task 10 — Management Insight Query (3 marks)

Write one additional SQL query that you believe would provide useful insight to BikeStores management using at least three tables.

Below the query, add a SQL comment of no more than three lines explaining:

1. the business question,
2. what the result measures, and
3. why management should care about it.

Marking Guide

Area	Marks	What is Assessed
Tasks 1–2: Core querying & aggregation	11	Joins, filtering, calculations, grouping and store metrics
Tasks 3–4: Business analysis	10	Aggregate comparison, inventory logic and correct result shaping
Tasks 5–6: Advanced analytics	12	Within-group ranking and month-over-month analysis
Task 7: View	4	Correct reusable customer summary and handling of customers without orders
Task 8: DML & transaction safety	4	Targeted update, validation and rollback
Task 9: Stored procedure	6	Parameters, date filtering, grouped result and error handling
Task 10: Independent insight	3	Useful multi-table query and concise business explanation

END OF ASSIGNMENT — TOTAL: 50 MARKS